

Exercise Sheet 3

Singular value decomposition (SVD)

Use of AI You may use AI tools to help draft code, but your final answers should show your own judgment and explain why the results look like that.

Setup in Python Use `numpy`, `pandas`, `matplotlib`, `Pillow`, and `scikit-learn`. Unless stated otherwise, use the following for SVD:

```
np.linalg.svd(X, full_matrices=False).
```

Notation For $\mathbf{X} \in \mathbb{R}^{n \times p}$, write its SVD using the r nonzero singular values as

$$\mathbf{X} = \sum_{j=1}^r \sigma_j \mathbf{u}_j \mathbf{v}_j^\top, \quad \sigma_1 \geq \cdots \geq \sigma_r > 0.$$

Problem 1

- Let $\mathbf{A} = \begin{pmatrix} 3 & 0 \\ 4 & 0 \end{pmatrix}$. Find the rank-one SVD by hand. Verify $\mathbf{A}\mathbf{v}_1 = \sigma_1\mathbf{u}_1$ and $\mathbf{A}^\top\mathbf{u}_1 = \sigma_1\mathbf{v}_1$.
- Let $\mathbf{B} = \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix}$. Compute its singular values. Use this example to explain why the spectral norm of a nonsymmetric matrix is its largest singular value.
- Starting from $\mathbf{C} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$ and orthogonal invariance, show that $\|\mathbf{C}\|_2 = \sigma_1(\mathbf{C})$, and $\|\mathbf{C}\|_F^2 = \sum_j \sigma_j^2(\mathbf{C})$.
- A matrix has nonzero singular values 8, 4, 1, 1. For $k = 2$, compute $\|\mathbf{X} - \mathbf{X}_k\|_F$, the retained squared-energy share, and the relative Frobenius error. State what the Eckart–Young–Mirsky theorem says about this error.

Problem 2 The file `E3/mona_lisa_public_domain.jpg` is a public domain image of *Mona Lisa*. Load it with `Pillow`, convert it to grayscale, resize it to height 360 pixels while preserving its aspect ratio, scale its intensities to $[0, 1]$. Let the resulting matrix be $\mathbf{X} \in \mathbb{R}^{n \times p}$.

- Compute n , p , $\text{rank}(\mathbf{X})$ numerically, and the intensity range. Display the image with a grayscale color map.
- Compute its SVD with `full_matrices=False`. Check numerically that the following ones are small:

$$\|\mathbf{U}^\top\mathbf{U} - \mathbf{I}\|_2, \quad \|\mathbf{V}^\top\mathbf{V} - \mathbf{I}\|_2, \quad \frac{\|\mathbf{X} - \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top\|_F}{\|\mathbf{X}\|_F}.$$

- c) Reconstruct and display $\mathbf{X}_k$ for $k \in \{1, 5, 10, 20, 50, 100\}$. Clip displayed intensities to $[0, 1]$, but compute all errors before clipping.
- d) For the same values of k , compute the actual Frobenius errors and compare them with $\left(\sum_{j>k} \sigma_j^2\right)^{1/2}$. Explain any discrepancy beyond floating point rounding.
- e) Find the smallest k that retains at least 90%, 95%, and 99% of the squared energy. For each value, compute the relative Frobenius error. Compare the mathematical criteria with your visual judgment.

Problem 3 The file `E3/market_data_2007_2025.csv` contains daily adjusted prices downloaded with `start_date= 2007-04-11` and `end_date= 2026-01-01` for SPY (S&P 500 equity market), QQQ (growth equities), TLT (long duration US Treasuries), HYG (high yield corporate credit), and VIX (option-implied S&P 500 volatility index). Use trading days only. Compute the five daily log changes $x_t^{(j)} = \log P_t^{(j)} - \log P_{t-1}^{(j)}$, and align them by date. Store the financial series in rows and dates in columns, as in Lecture 3: $\mathbf{R} \in \mathbb{R}^{5 \times T}$, $R_{jt} = x_t^{(j)}$, where T is the number of aligned trading dates. The five rows are SPY, QQQ, TLT, HYG, and VIX, in that order. The CSV has dates in rows: transpose it before computing the SVD.

- a) Compute the SVD $\mathbf{R} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$. State what the left singular vector $\mathbf{u}_1 \in \mathbb{R}^5$ and the right singular vector $\mathbf{v}_1 \in \mathbb{R}^T$ represent. Which of $\mathbf{v}_1$ and $\sigma_1 \mathbf{v}_1$ gives the actual dated coefficients of the first asset pattern?
- b) Plot the singular values and cumulative squared-energy shares for $\mathbf{R}$. Treat the singular-value curve as a scree plot. State whether it has a convincing elbow and give a defensible elbow-based k , if one exists. For use in part c), orient the leading left singular vector so that its SPY coefficient is positive, and report its coefficients. If you reverse its sign, also reverse the sign of the paired right singular vector. For the first date, calculate the rank-one reconstructed column $\sigma_1(v_1)_1 \mathbf{u}_1$ and interpret its entries.
- c) Let $\mathbf{u}_1$ be the oriented leading left singular vector from part c), and let $\mathcal{T} = \{\text{SPY, QQQ, TLT, HYG}\}$ and $\mathcal{I}_+ = \{j \in \mathcal{T} : u_{j1} > 0\}$. For $j \in \mathcal{I}_+$, set $w_j = \frac{u_{j1}}{\sum_{\ell \in \mathcal{I}_+} u_{\ell 1}}$. Compute $\mathcal{I}_+$ and the weights. Convert the log changes to simple returns $q_t^{(j)} = \exp(x_t^{(j)}) - 1$. Plot the cumulative return

$$\prod_{s \leq t} \left(1 + \sum_{j \in \mathcal{I}_+} w_j q_s^{(j)} \right) - 1$$

beside that of a daily equally weighted portfolio of the same ETFs.

- d) For $k = 1, 2, 3, 4$, reconstruct $\mathbf{R}_k$. Which rank retains at least 90% of the squared energy?
- e) For $k = 2$, plot the original changes and their reconstruction over the final 250 trading days. Plot the residuals separately. Describe what is retained and what is lost.
- f) The portfolio weights in part c) were fitted using the same dates on which performance is plotted. Explain why neither a large singular value nor a higher terminal cumulative return establishes a profitable future strategy.